IC 272: DATA SCIENCE - III
LAB ASSIGNMENT – III
Data visualization and statistics from data

Student's Name: Abhay Gupta **Mobile No:** 9511334630

Roll Number: B20075 **Branch:**CSE

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Table 1 Mean, median, mode, minimum, maximum and standard deviation for all the attributes

S. No.	Attributes	Mean	Median	Mode	Min.	Max.	S.D.
1	pregs	3.845	3	1	0	17	3.369
2	plas	120.894	117	99	0	199	31.972
3	pres (in mm Hg)	69.105	72	70	0	122	19.355
4	skin (in mm)	20.536	23	0	0	99	15.952
5	test (in mu U/mL)	79.799	30.5	0	0	846	115.244
6	BMI (in kg/m^2)	31.992	32	32	0	67.10	7.884
7	pedi	0.471	0.372	0.254	0.078	2.42	0.331
8	Age (in years)	33.240	29	22	21	81	11.760

Inferences:

1. From the above data, we can see that whenever standard deviation is close to zero, the mean, median and mode becomes approximately equal.
2. As the standard deviation increases, the range increases. (where, Range = max-min)

2 a.

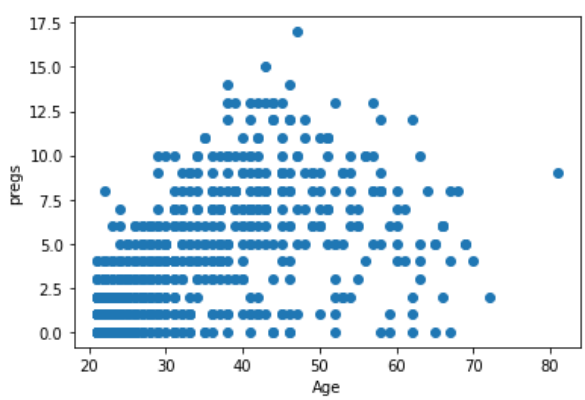

Figure 1 Scatter plot: Age (in years) vs. pregs

Inferences:

1. Attribute age has a moderate positive correlation with pregs.
2. Scatter plot is dense in the age of 20 to 40.
3. With increase in age, pregs will increase.

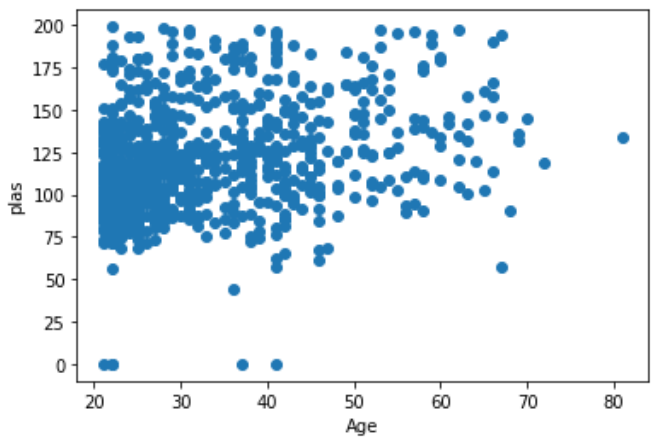

Figure 2 Scatter plot: Age (in years) vs. plas

Inferences:

1. Attribute age has a weak positive correlation with plas.
2. Density decreases as age increases in the above scatter plot.
3. With increase in age, plas will increase.

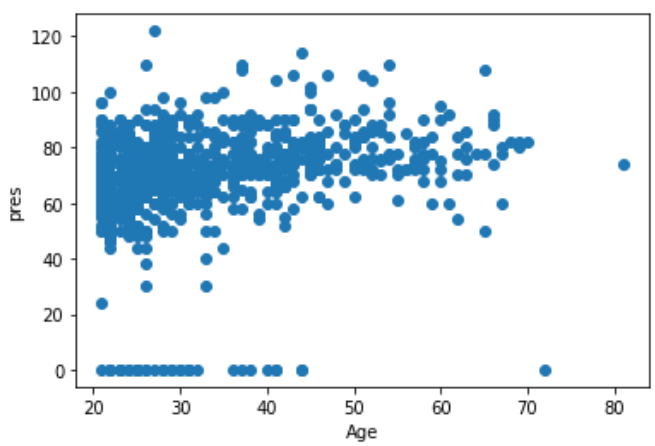

Figure 3 Scatter plot: Age (in years) vs. pres (in mm Hg)

Inferences:

1. Attribute age has a weak positive correlation with pres.
2. Density decreases as age increases.
3. With increase in age, pres will increase.

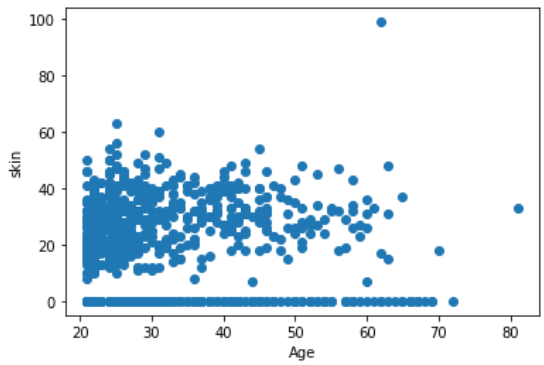

Figure 4 Scatter plot: Age (in years) vs. skin (in mm)

Inferences:

1. With increase in age the skin attribute does not change it's value significantly. So, we can say they are not correlated or very weakly correlated.
2. Density decreases as age increases.
3. There are some outliers in the above graph.

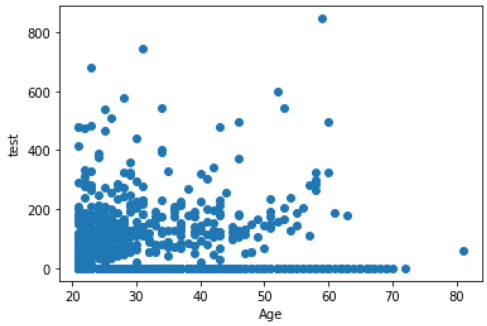

Figure 5 Scatter plot: Age (in years) vs. test (in mm U/mL)

Inferences:

1. With increase in age attribute, the test attribute increases here.
2. Density of scatter plot decreases as the value of age increases.

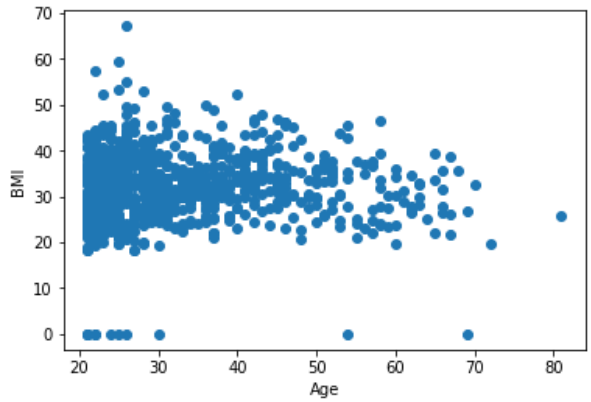

Figure 6 Scatter plot: Age (in years) vs. BMI (in kg/m^2)

Inferences:

1. With increase in age the BMI attribute does not change it's value significantly. So, we can say they are not correlated or very weakly correlated.
2. Density decreases as age increases.

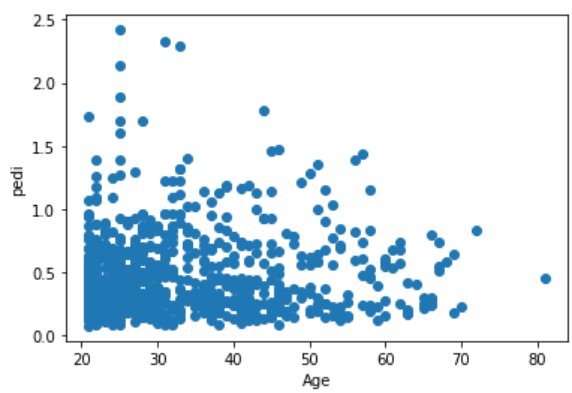

Figure 7 Scatter plot: Age (in years) vs. pedi

Inferences:

1. There is very little increase in the pedi attribute, as age increases. So, we can say that they are weakly correlated.
2. Density of the scatter plot decreases as age increases.
3. There are some outliers in the above scatter plot.

b.

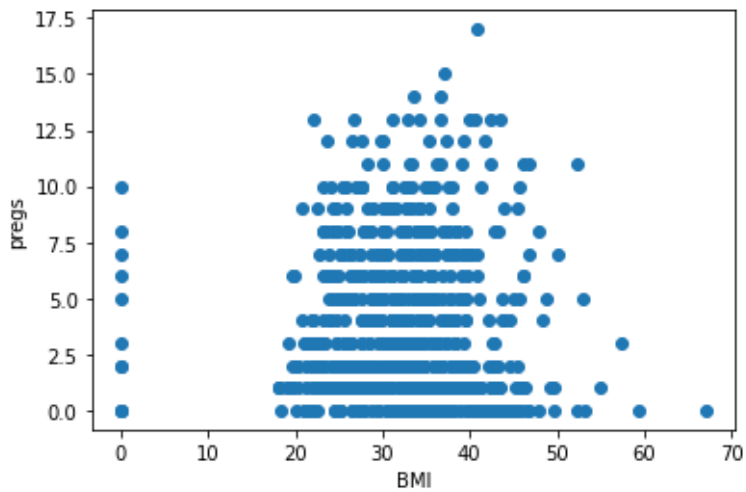

Figure 8 Scatter plot: BMI (in kg/m²) vs. pregs

Inferences:

1. There is no significant change in pregs as we increase the BMI attribute
2. The scatter plot is denser between the age of 20 to 55.

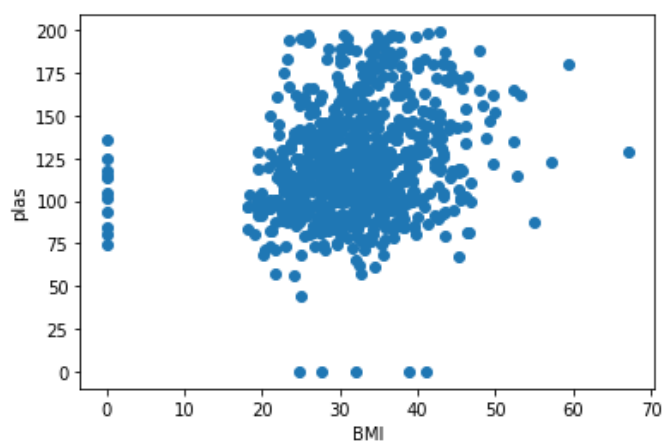

Figure 9 Scatter plot: BMI (in kg/m²) vs. plas

Inferences:

1. As BMI increases, there is little increase in the attribute plas. So, we can say that they are weakly correlated.
2. Majority of the points lie between age of 20 to 40.

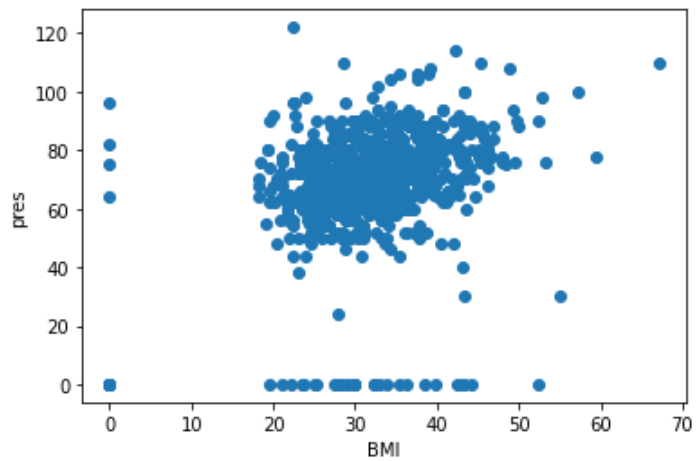

Figure 10 Scatter plot: BMI (in kg/m²) vs. pres (in mm Hg)

Inferences:

1. As BMI increases, there is little increase in the attribute pres. So, we can say that they are weakly correlated .
2. Majority of the points lie between age of 20 to 45.

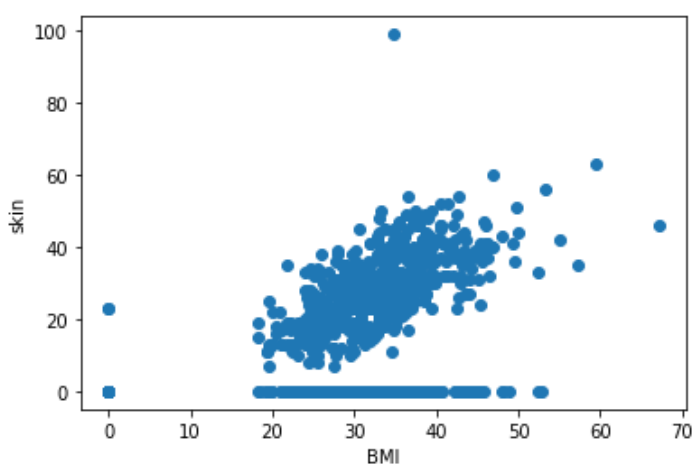

Figure 11 Scatter plot: BMI (in kg/m²) vs. skin (in mm)

Inferences:

1. As BMI increases, skin attribute also increases in the above plot.
2. Majority of the points lie between age of 20 to 45.

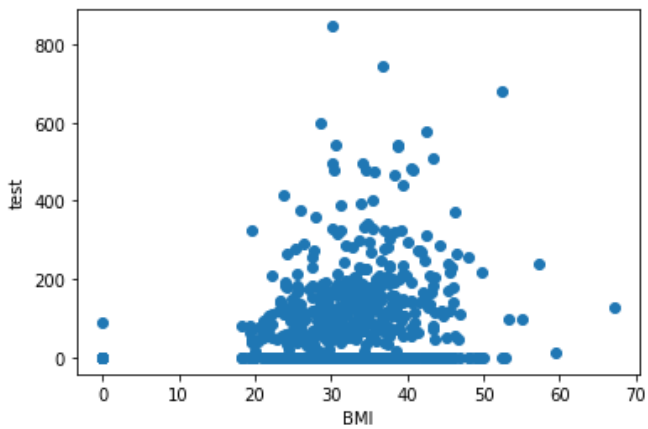

Figure 12 Scatter plot: BMI (in kg/m²) vs. test (in mm U/mL)

Inferences:

1. We can not observe any significant pattern on the increase of BMI.
2. Majority of the points lie between age of 20 to 45.

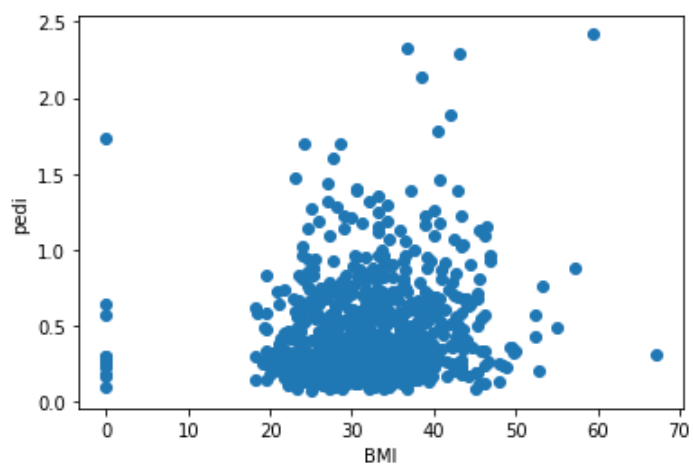

Figure 13 Scatter plot: BMI (in kg/m²) vs. pedi

Inferences:

1. We can not observe any significant pattern on the increase of BMI.
2. Majority of the points lie between age of 20 to 45.

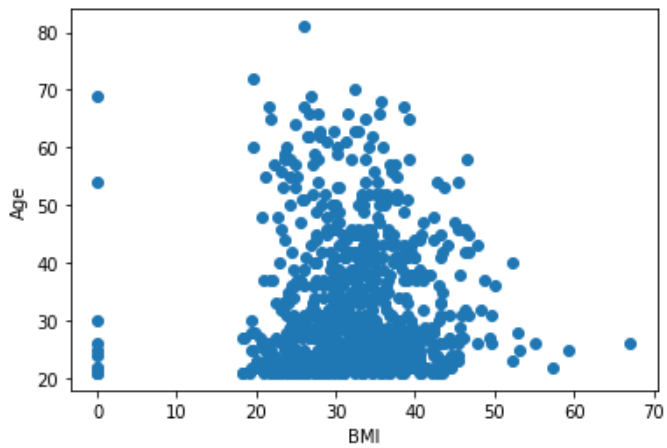

Figure 14 Scatter plot: BMI (in kg/m^2) vs. Age (in years)

Inferences:

1. We can not observe any significant pattern on the increase of BMI.
2. Majority of the points lie between age of 20 to 45.

3 a.

Table 3 Correlation coefficient value computed between age and all other attributes

S. No.	Attributes	Correlation Coefficient Value
1	pregs	0.544
2	plas	0.263
3	pres (in mm Hg)	0.239
4	skin (in mm)	-0.113
5	test (in mu U/mL)	-0.042
6	BMI (in kg/m^2)	0.036
7	pedi	0.033
8	Age (in years)	1

Inferences:

1. There is a moderate positive correlation between age and pregs attribute. Plas and pres attributes have weak positive correlation with age. BMI and pedi have very weak positive correlation (almost no correlation) with age. Skin and test attributes have very weak negative correlation with age.
2. With increase in age attribute plas, pres, BMI, pedi attributes will increase. Whereas, skin and test will decrease.
3. The strength of correlation depicted from the above correlation coefficients is almost similar to that observed from the scatter plots above.

b.

Table 4 Correlation coefficient value computed between BMI and all other attributes

S. No.	Attributes	Correlation Coefficient Value
1	pregs	0.017
2	plas	0.221
3	pres (in mm Hg)	0.281
4	skin (in mm)	0.392
5	test (in mu U/mL)	0.197
6	BMI (in kg/m^2)	1
7	pedi	0.140
8	Age (in years)	0.036

Inferences:

1. Skin, pres and plas attributes have weak positive correlation with BMI. All other attributes have very weak positive correlation (almost no correlation) with BMI.
2. With increase in age, all the attributes will increase.
3. The strength of correlation depicted from the above correlation coefficients is almost similar to that observed from the scatter plots above.

4 a.

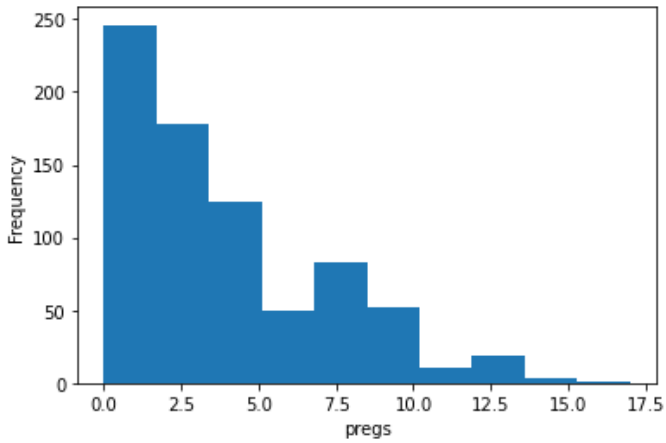

Figure 15 Histogram depiction of attribute pregs

Inferences:

1. From the histogram we can see that, as the attribute pregs is increasing, the frequency is decreasing. Maximum frequency is close to 250 then it comes to 175 (approx.) and then comes close to 125. Further it goes to 50 then to 75 and then again to 50.
2. From the histogram, we can see that maximum frequency lies between 0-2.5. So, the mode will lie in the range of 0-2.5.

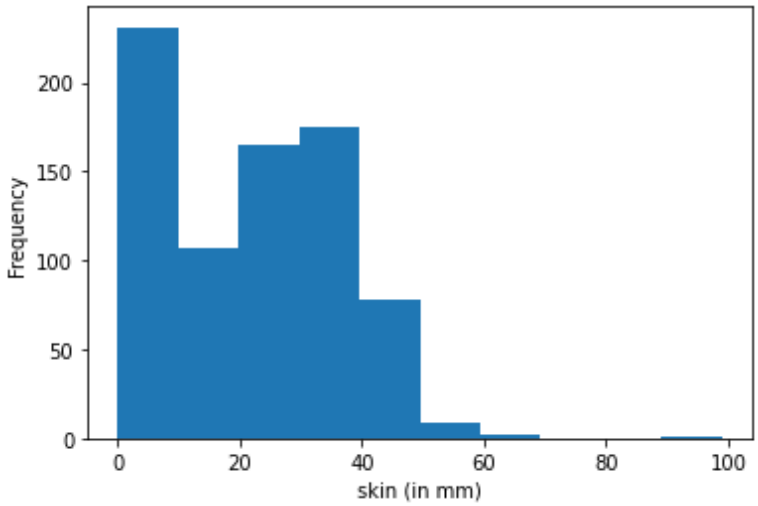

Figure 16 Histogram depiction of attribute skin

Inferences:

1. Attribute skin has a frequency close to 230 for the range of 0-10. Then it goes to 100. For the range 20-30 frequency is 160 (approx.) then it goes to 175. For the range 40-50 the frequency is 75. Then it is close to 20.
2. From the histogram, we can see that maximum frequency lies between 0-10. So, the mode will lie in the range of 0-10.

5

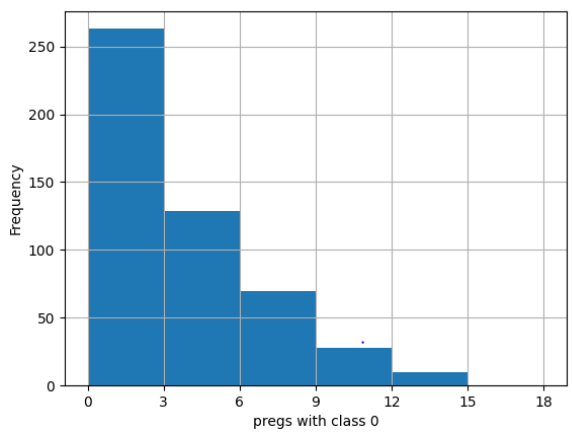

Figure 17 Histogram depiction of attribute pregs for class 0

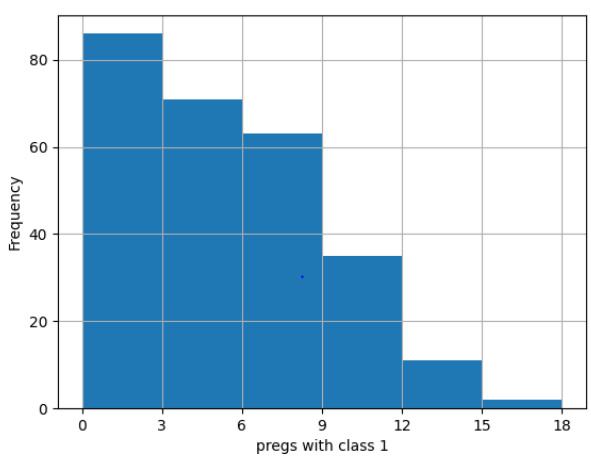

Figure 18 Histogram depiction of attribute pregs for class 1

Inferences:

1. From the histogram, mode of the attribute pregs for class 0 lies in the range of 0-1. For class 1 the mode lies in the range of 0-1.25.
2. (a). For the bin 0-3, frequency with class 0 is 263 and with class 1 it is 86.
   (b). For 3-6, frequency with class 0 is 263 and with class 1 it is 86.
   (c). For 6-9, frequency with class 0 is 69 and with class 1 it is 63.
   (d) For 9-12, frequency with class 0 is 28 and with class 1 it is 35.
   (e) For 12-15, frequency with class 0 is 10 and with class 1 it is 11.
   (f) For 15-18,frequency with class 0 is 0 and with class 1 it is 2.

6

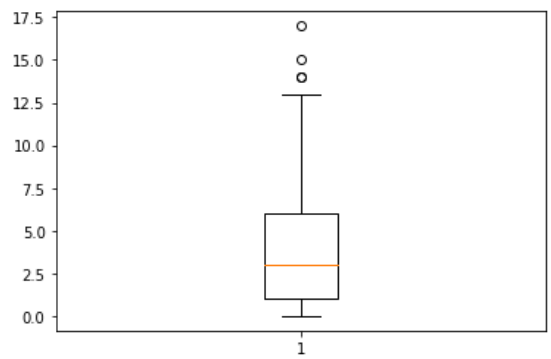

Figure 19 Boxplot for attribute pregs

Inferences:

1. There are 3 outliers and their values are 14, 15 and 17
2. The Inter quartile range is 5
3. The data varies from 0 to 17. It's bottom whisker is 1 and top whisker is 6. It's median is 3
4. It is positively skewed.
5. The maximum value, minimum value, median matches with that of question 1.

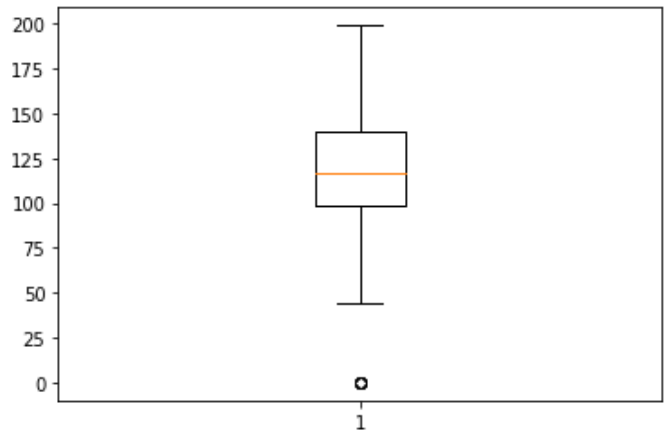

Figure 20 Boxplot for attribute plas

Inferences:

1. There is one outlier whose value is 0.
2. The Inter quartile range is 41.
3. The data varies from 0 to 199.It's bottom whisker is 44 and top whisker is 199. It's median is 117.
4. It is positively skewed.
5. The maximum value, minimum value, median matches with that of question 1.

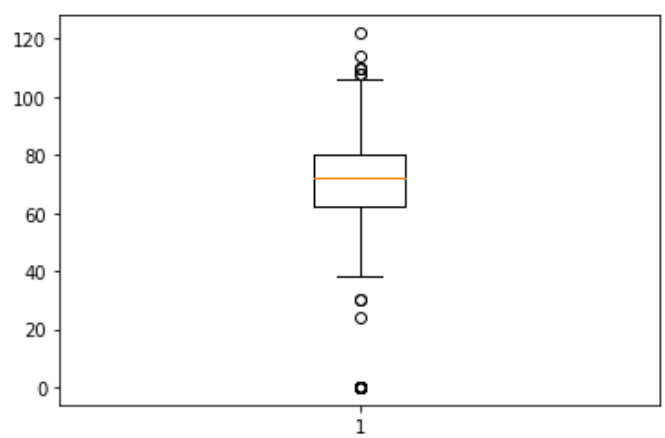

Figure 21 Boxplot for attribute pres(in mm Hg)

Inferences:

1. There are some outliers whose values are 107,110,114,122,0,24,30.
2. The Inter quartile range is 18
3. The data varies from 0 to 122.It's bottom whisker is 38 and top whisker is 106. It's median is 72
4. It is negatively skewed.
5. The maximum value, minimum value, median matches with that of question 1.

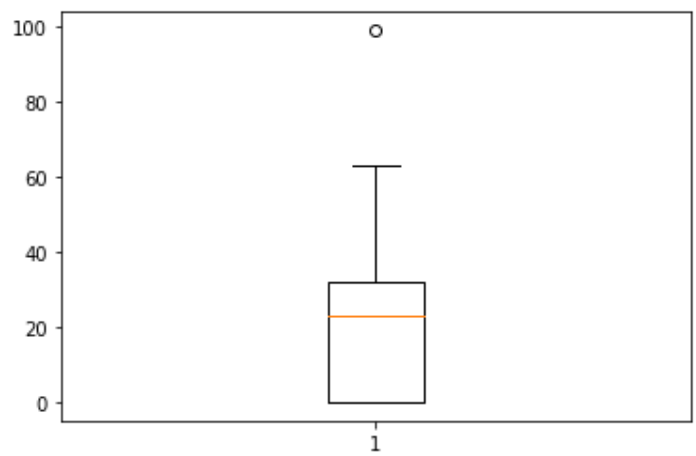

Figure 22 Boxplot for attribute skin(in mm)

Inferences:

1. There is one outlier whose value is 99.
2. The Inter quartile range is 32
3. The data varies from 0 to 99. It's bottom whisker is 0 and top whisker is 63. It's median is 23
4. It is negatively skewed.
5. The maximum value, minimum value, median matches with that of question 1.

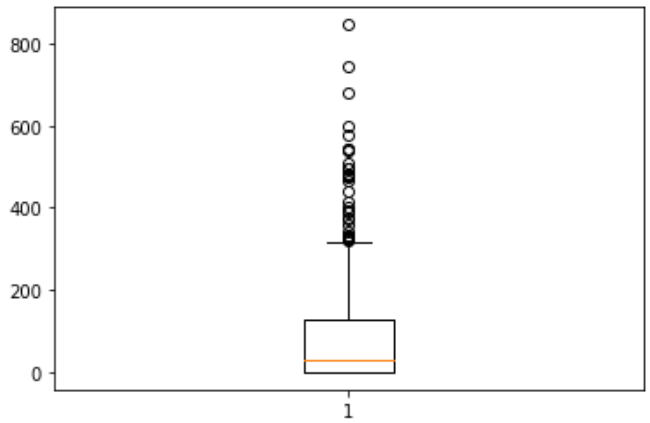

Figure 23 Boxplot for attribute test (mu U/mL)

Inferences:

1. There are many outliers which can be seen in the box plot. All outliers lie above the top whisker.
2. The Inter quartile range is 128
3. The data varies from 0 to 847.It's bottom whisker is 0 and top whisker is 318. It's median is 30.5
4. It is positively skewed.
5. The maximum value, minimum value, median matches with that of question 1.

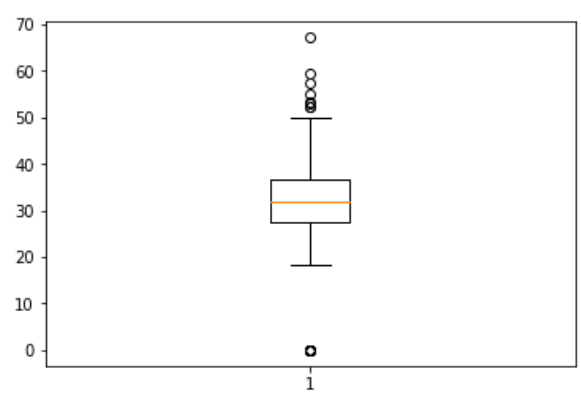

Figure 24 Boxplot for attribute BMI (in kg/m^2)

Inferences:

1. There are many outliers which can be seen in the box plot. All outliers lie above the top whisker
2. The Inter quartile range is 18
3. The data varies from 0 to 67. It's bottom whisker is 18 and top whisker is 50. It's median is 32
4. This data seems to be unskewed as median lies close to the mid of IQR.
5. The maximum value, minimum value, median matches with that of question 1.

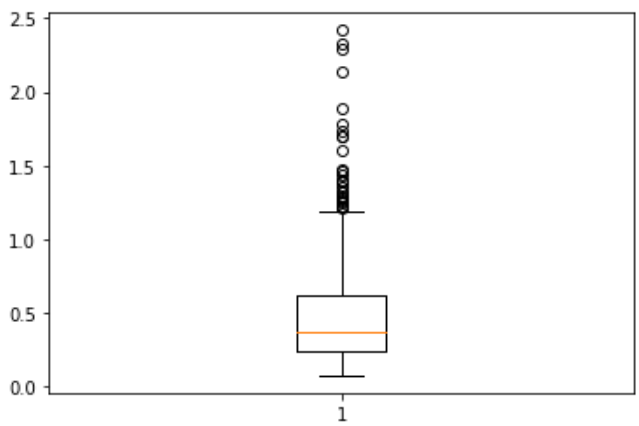

Figure 25 Boxplot for attribute pedi

Inferences:

1. There are many outliers which can be seen in the box plot. All outliers lie above the top whisker
2. The Inter quartile range is 0.4
3. The data varies from 0.078 to 2.41. It's bottom whisker is 0.078 and top whisker is 1.17. It's median is 0.372
4. This data is positively skewed.
5. The maximum value, minimum value, median matches with that of question 1.

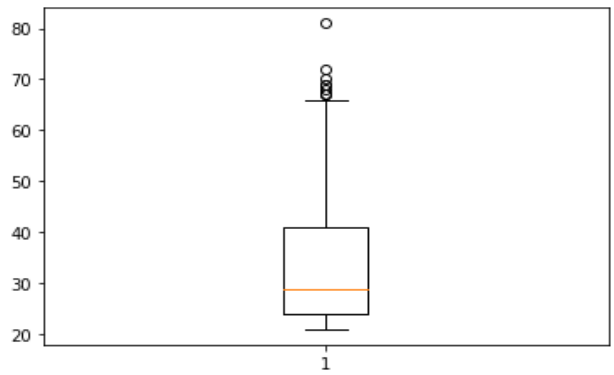

Figure 26 Boxplot for attribute Age (in years)

Inferences:

1. There are many outliers which can be seen in the box plot. All outliers lie above the top whisker
2. The Inter quartile range is 17
3. The data varies from 21 to 81. It's bottom whisker is 21 and top whisker is 66. It's median is 29.
4. This data is positively skewed.
5. The maximum value, minimum value, median matches with that of question 1.